



Syntax II

Merlin Balkhash
Department of Linguistics, Rutgers University
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Please pick up your **name card** from the front

Words and Phrases

- **parts of speech** are the basic building blocks of **phrases**
- a sentence is a also kind of phrase (**tense phrase**, TP)
- sentence may contain a noun and a verb:

Merlin ran.
N V

- however, there are also other things that can go in the same spot as this **N**:

The thief ran.
D N V

The teacher from chicago ran.
D N P N V



Distribution

when you figure out the syntax
and lexicon but not the
morphology



- these ‘chunks’ also have other similar **distributional** properties:
 - they only appear in certain context or environment
- in other words, they appear in **the same place** and they accomplish **a similar purpose**

Roger.

who ran? *The thief.*

The teacher from Chicago.

It is Roger who ran.

It is the thief who ran.

It is the teacher from Chicago who ran.

Constituents: Intuition



- imagine that you want to embed a [weblink](#) in the following sentence:
 - *The cat loves **his** toy.*
 - *The cat **loves his** toy.*
 - ***The cat** loves his toy.*
 - *The **cat loves** his toy.*
- which of these options above would be **natural** choices?
- do any of them strike you as **unnatural** or **unlikely** choices?
- although a sentence is composed of separate words, it is **NOT** constructed like a **parachute**, where each word independently attaches to the sentence as a whole
- instead, words first group into component parts that function **together as units**, and these units then combine to form the sentence

Constituents

constituent: a group of words that **acts together** as a **unit**

- consider the following sentences and find out the **subject** for each sentence?

1. *Francisco* ***studies linguistics.***
2. *The tall boy* ***studies linguistics.***
3. *The tall boy from New Jersey* ***studies linguistics.***
4. *The tall boy named Francisco* ***studies linguistics.***
5. *The tall boy that I met last week* ***studies linguistics.***

large phrases and sentences



phrases



words



morphemes

(a video on constituent)

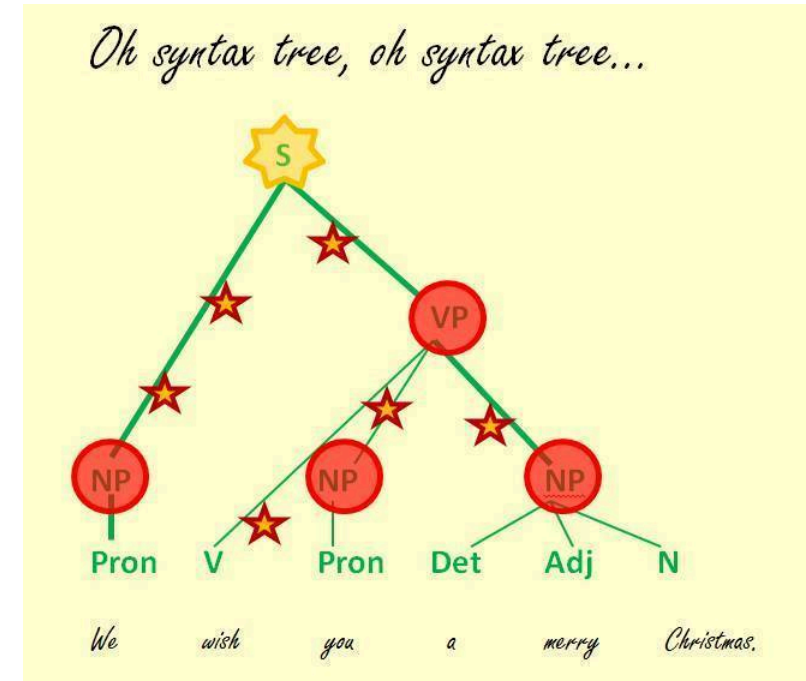
Constituents

constituent: a group of words that **acts together** as a **unit**

- consider the following sentences and find out the **subject** for each sentence?

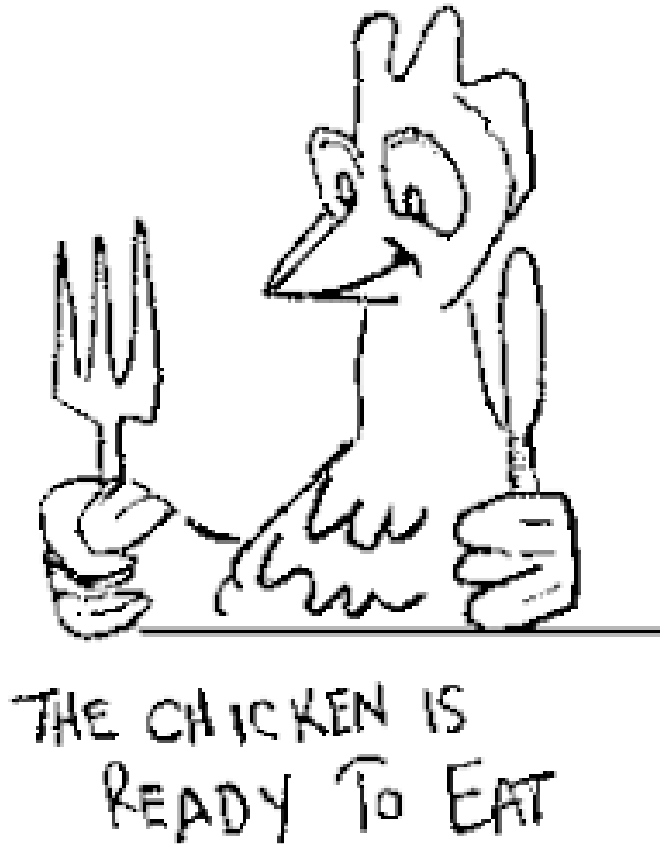
- Francisco **studies linguistics.**
- The tall boy **studies linguistics.**
- The tall boy from New Jersey **studies linguistics.**
- The tall boy named Francisco **studies linguistics.**
- The tall boy that I met last week **studies linguistics.**

- SUBJECT can be **one** or **many** words, but they **act together** to describe a single concept



(a video on constituent)

Back to the Ambiguity Case AGAIN



- remember the case of structural ambiguity:

Arthur read the book on the desk.

- this type of ambiguity actually reveals that there is more than one **constituent organisation** for this sentence:
 - Arthur read *[the book] [on the desk]*.
 - the green part is linked to the verb 'read' directly, meaning 'read on the desk'
 - Arthur read *[the book on the desk]*.
 - the green part is linked to the noun 'book' directly, meaning 'book on the desk'

- speakers are aware of this fact

Phrases: Types of Constituents

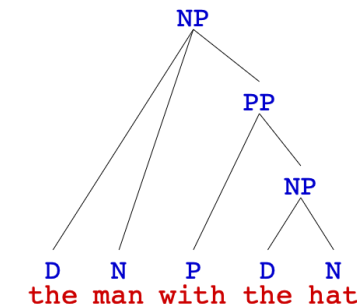
- each phrase has a main word aka **head**
- phrases are named after the **head** word's **part of speech**:

- **noun phrase (NP)**: man, the man, the happy man, the man with the hat
- **prepositional phrase (PP)**: into the forest, by the author, for good luck
- **verb phrase (VP)**: eats, eats pizza, eats quickly, eats pizza quickly at noon
- **adjective phrase (AdjP)**: happy, ugly

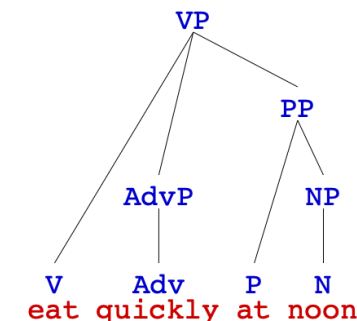
- **adverb phrase (AdvP)**: quickly, happily

- **one constituent can contain another:**

- the man with the hat (**NP**) contains with the hat (**PP**), which contains the hat (**NP**)

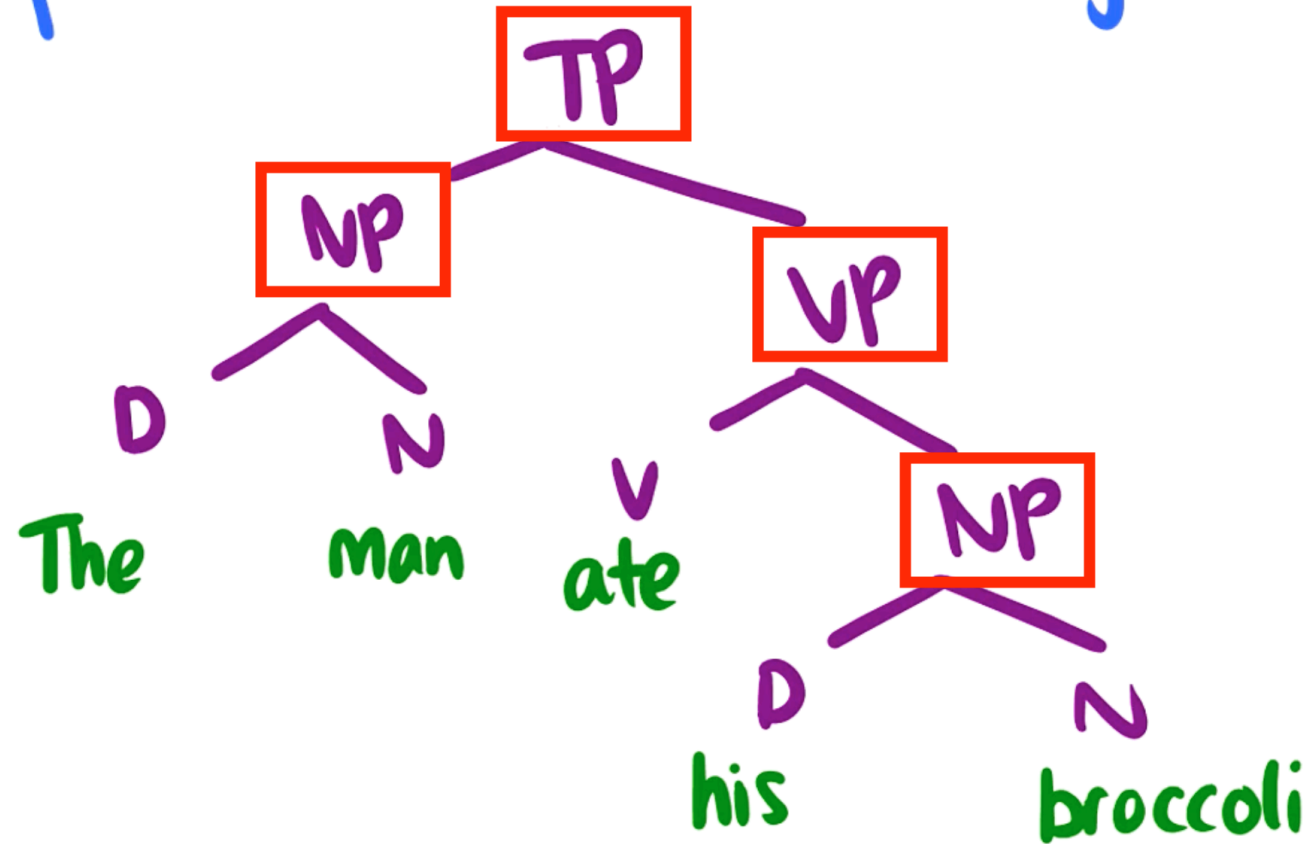


- eats quickly at noon (**VP**) contains quickly (**AdvP**) and at noon (**PP**)



Constituents

- A group of words that function together as a unit.



$[_{TP}[_{NP}[_{D} \text{The}][_N \text{man}]][_{VP}[_{V} \text{ate}][_{NP}[_{D} \text{his}][_N \text{broccoli}]]]]]$

Constituency Tests

- a **constituency test** applies to a portion of a sentence and identify **whether** groups of words act together as a unit or not
- if the test doesn't apply, we can still use it but using * (**ungrammatical**) notation to show that the test doesn't work
- **four** types of test:
 - **replacement/substitution** test
 - **movement** test
 - **fragment/stand-alone** test
 - **coordination** test
- **these don't work 100% of the time**, and some tests work better with some sentences
 - hence, you might have to try several



***[The lugubrious man]** is eating **[his bland meal]**.*

Constituency Tests : Procedure Passed

1. begin with **Sentence A**, with a sequence to be tested (usually underlined)

Sentence A: The anonymous girl escaped.

2. use **some test(s)** to create a new sentence: **Sentence B**

Sentence B: It is the anonymous girl who escaped.

3. decide if **Sentence B** is syntactically **well-formed**



if **Sentence B** is grammatical, then the
sequence in **Sentence A** **passed** this test

the sequence is **a constituent**

if Sentence B is ungrammatical, then the
sequence in Sentence A **failed** this test

the sequence is **NOT a constituent**

Constituency Tests : Procedure Failed

1. begin with **Sentence A**, with a sequence to be tested (usually underlined)

Sentence A: *The anonymous girl escaped.*

2. use **some test(s)** to create a new sentence: **Sentence B**.

Sentence B: ** It is anonymous girl who escaped.*

3. decide if **Sentence B** is syntactically **well-formed**



if Sentence B is grammatical, then the
sequence in Sentence A **passed** this test

the sequence is **a constituent**

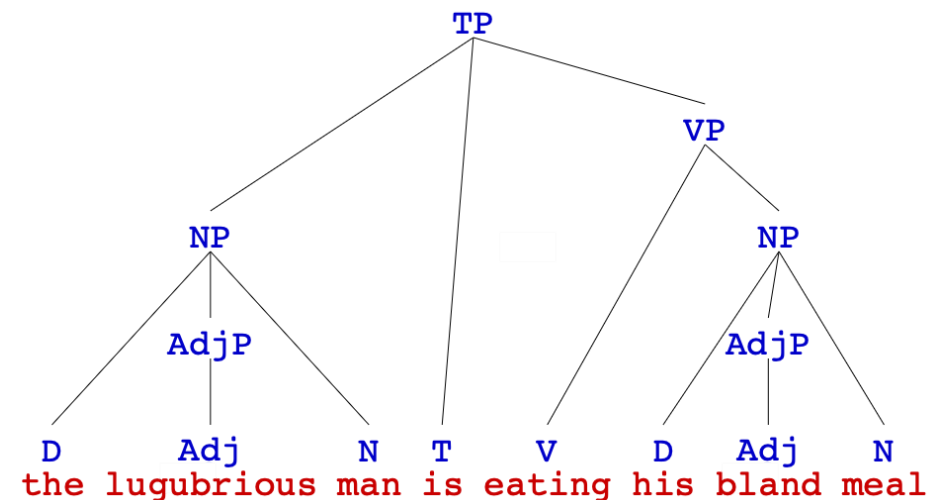
if **Sentence B** is ungrammatical, then the
sequence in **Sentence A** **failed** this test

the sequence is **NOT a constituent**

Replacement (Substitution) Test

[The lugubrious man] is eating *[his bland meal]*.

- **replacement test**: replace a group of words with a SINGLE **pro-form**
- if it works, they are one constituent
- while applying replacement tests, it is important to replace the **whole string** under investigation
 - do **NOT** include anything more
 - do **NOT** miss anything
 - do **NOT** paraphrase the sentence
 - apply the test to the **whole** and **only** the string under investigation

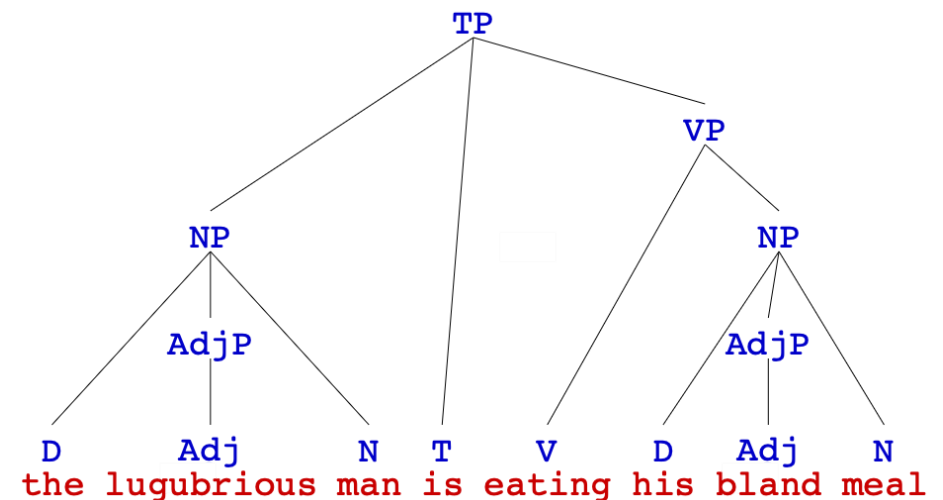


'The lugubrious man is eating his bland meal.'

Replacement (Substitution) Test

[The lugubrious man] is eating *[his bland meal]*.

- **replacement test**: replace a group of words with a SINGLE **pro-form**
- **pro-form**: pronoun or other generic expression like there, like that, do, do that, that way
 - a **noun-phrase-like** constituent can be replaced by a **pronoun** (it, them, ...)
 - a **verb-phrase-like** constituent can be replaced by **do/does/did/did** (so/that)
 - a **preposition-phrase-like** constituent can be replaced by **here/there**

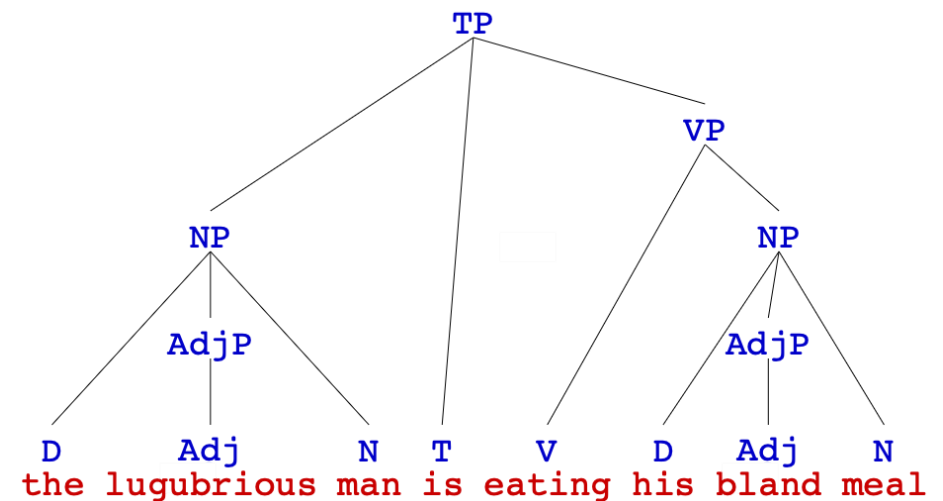


'The lugubrious man is eating his bland meal.'

Replacement (Substitution) Test

[The lugubrious man] is eating *[his bland meal]*.

- **replacement test**: replace a group of words with a SINGLE **pro-form** with a SINGLE **pro-form**
- **pro-form**: pronoun or other generic expression like there, like that, do, do that, that way.
- *The lugubrious man* is eating his bland meal.
 - replacing 'the lugubrious man' with 'he'
 - → *He* is eating his bland meal. ✓
- *The lugubrious man* *is eating his bland* meal.
 - a **SINGLE pro-form** cannot be found
 - → *The lugubrious man* *???* meal. ✗

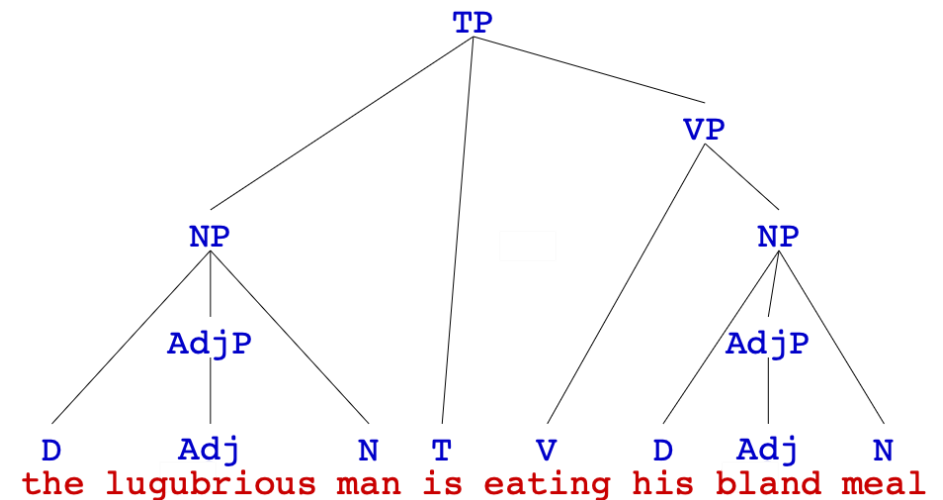


'The lugubrious man is eating his bland meal.'

Replacement (Substitution) Test

[The lugubrious man] is eating *[his bland meal]*.

- **replacement test**: replace a group of words with a SINGLE **pro-form**
 - **pro-form**: pronoun or other generic expression like there, like that, do, do that, that way.
 - try the **replacement test** with following cases
1. *The lugubrious man is eating his bland meal.*
 2. *The lugubrious man is eating his bland meal.*
 3. *The lugubrious man is eating his bland meal.*

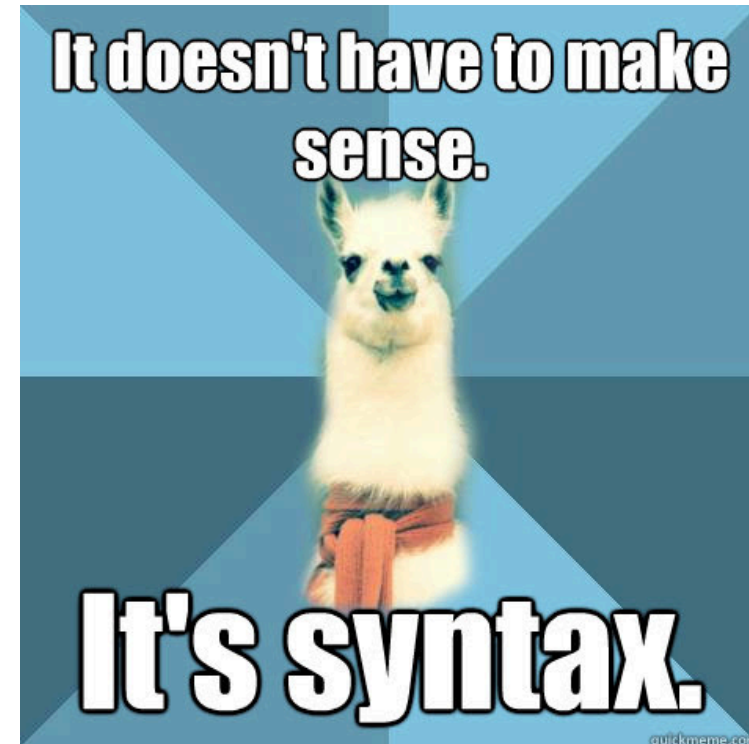


'The lugubrious man is eating his bland meal.'

Exercise I: Replacement (Substitution) Test

try to apply the **replacement test** to the underlined string and determine whether it is a **constituent**

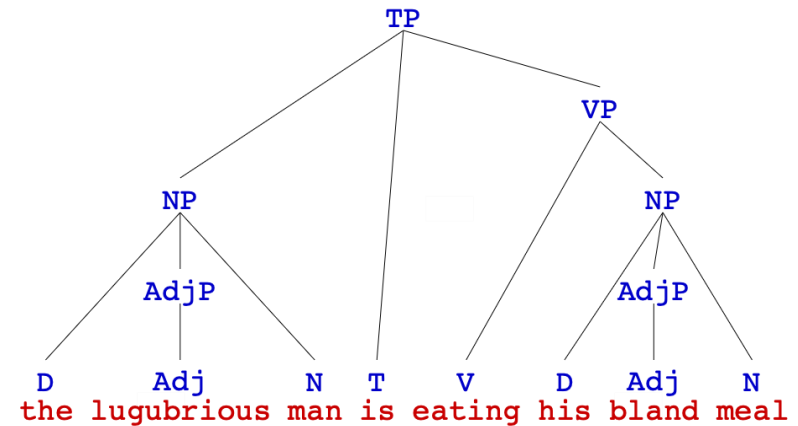
1. *I adopted the dog from Alaska.*
2. *A tall giraffe walked by.*
3. *The box of snacks fell off the table.*
4. *Merlin likes to sing in the shower.*
5. *Arthur read the book on the table.*



Movement Test

[The lugubrious man] is eating *[his bland meal]*.

- **movement test**: moving a string of words together to a **different part** of the sentence and still meaning the **same thing**
- there are two types of **movement test**
- **preposing**: putting the string of words before a **is was/are what** or **is was/are who** at the beginning of the sentence
- The lugubrious man is eating the bland meal.
→ The lugubrious man
 is who is eating the bland meal.

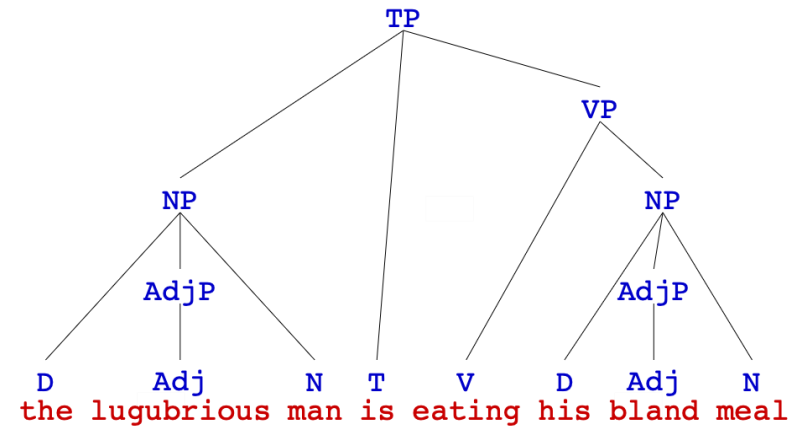


'The lugubrious man is eating his bland meal.'

Movement Test

[The lugubrious man] is eating *[his bland meal]*.

- **movement test**: moving a string of words together to a **different part** of the sentence and still meaning the **same thing**
- there are two types of **movement test**
- **clefting**: putting the string of words between **it was/it is** and a **that/who** at the beginning of the sentence
- *The lugubrious man is eating the bland meal.*
→ *It is the bland meal*
that the lugubrious man is eating.



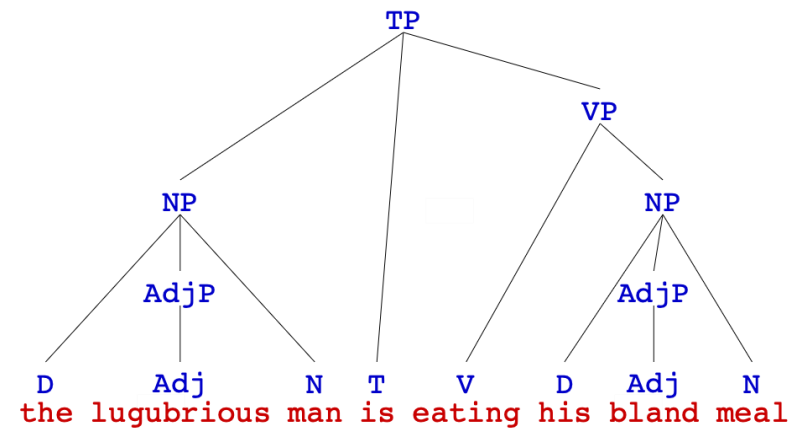
'The lugubrious man is eating his bland meal.'

Movement Test

[The lugubrious man] is eating *[his bland meal]*.

- **movement test**: moving a string of words together to a **different part** of the sentence and still meaning the **same thing**
- **preposing**: ... **is/was/are/were what...**
- **clefting**: **it was/is ... that/who**
- try the **movement test** with following cases

1. *The lugubrious man is eating his bland meal.*
2. *The lugubrious man is eating his bland meal.*
3. *The lugubrious man is eating, his bland meal.*

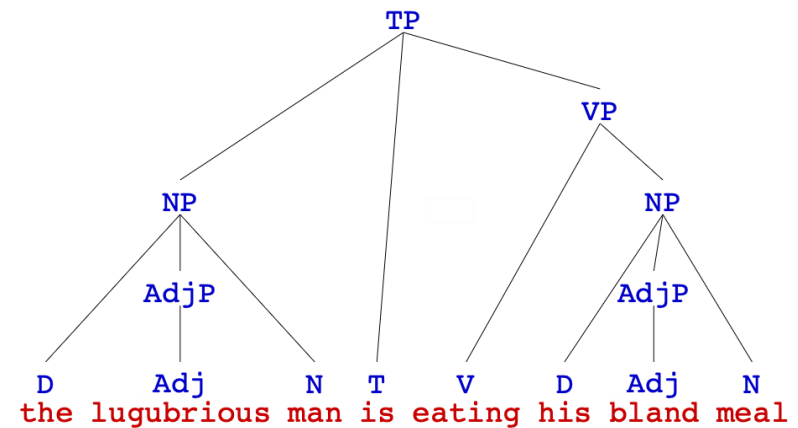


'The lugubrious man is eating his bland meal.'

Fragment (Stand-Alone) Test

[The lugubrious man] is eating *[his bland meal]*.

- **stand alone test**: come up with a question that can be **answered** by just saying the group of words you want to test **on its own**
- again, use the entire string under investigation
 - do **NOT** include anything more
 - do **NOT** miss anything
- *The lugubrious man* is eating his bland meal.
→ *Who is eating his bland meal?*
The lugubrious man.
- *The lugubrious man* is *eating his bland* meal.
→ * *The lugubrious man is what meal?*
ting his bland.



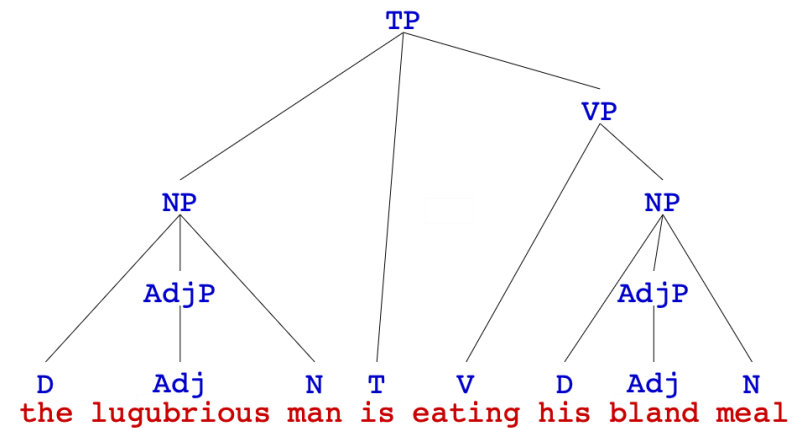
'The lugubrious man is eating his bland meal.'

Fragment (Stand-Alone) Test

[The lugubrious man] is eating *[his bland meal]*.

- **stand alone test**: come up with a question that can be **answered** by just saying the group of words you want to test **on its own**
- **constituents** can often stand alone and act as the answers or as the content of a question
- try the **fragment test** with following cases

1. *The lugubrious man is eating his bland meal.*
2. *The lugubrious man is eating his bland meal.*
3. *The lugubrious man is eating his bland meal.*

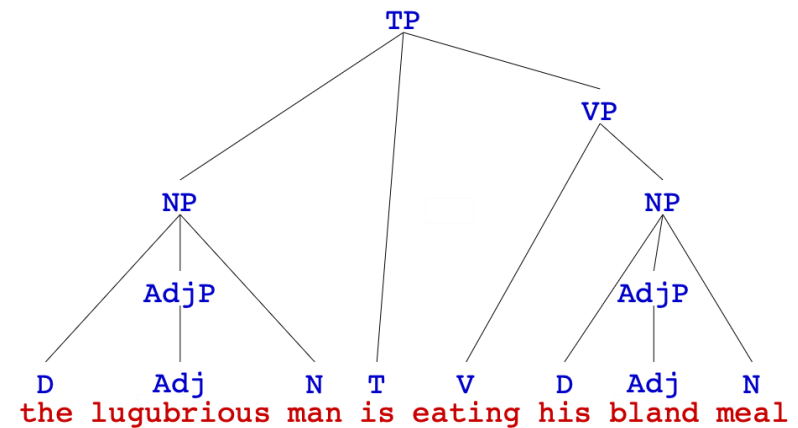


'The lugubrious man is eating his bland meal.'

Coordination Test

[The lugubrious man] is eating *[his bland meal]*.

- **coordination test**: coordinating the tested string with another string of the same type
- a constituent of type **X** can be coordinated with another constituent of the same type
 - coordinating conjunctions: **and** / **or**
- a **noun-phrase-like** string is coordinated with another **noun phrase**
 - the most safe choice would be a **pronoun** or a **common noun** (a name)
- a **prepositional-phrase-like** string is coordinated with another **prepositional phrase**
 - typically with **here/there**

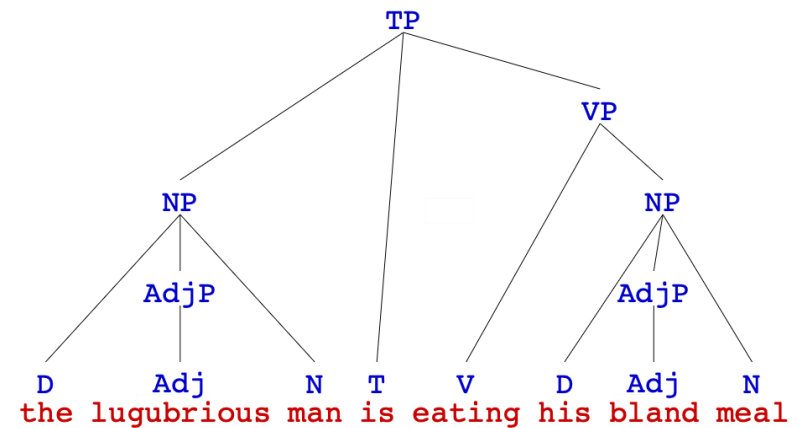


'The lugubrious man is eating his bland meal.'

Coordination Test

[The lugubrious man] is eating *[his bland meal]*.

- **coordination test:** coordinating the tested string with another string of the same type
- a constituent of type **X** can be coordinated with another constituent of the same type
 - coordinating conjunctions: **and** / **or**
- The lugubrious man is eating his bland meal.
 → The lugubrious man or Rumpelstilzchen is eating his bland meal.
- The lugubrious man is eating his bland meal.
 → The lugubrious man is eating his bland meal and running.



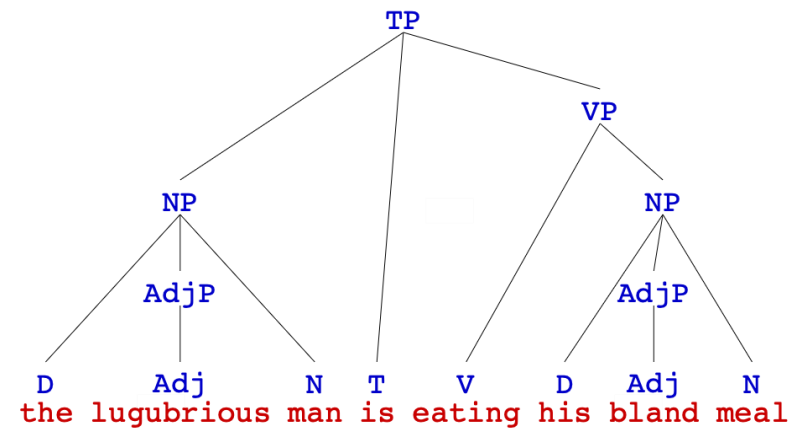
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Coordination Test

[The lugubrious man] is eating *[his bland meal]*.

- **coordination test**: coordinating the tested string with another string of the same type
- a constituent of type **X** can be coordinated with another constituent of the same type
 - coordinating conjunctions: **and** / **or**
- try the **coordination test** with following cases

1. *The lugubrious man is eating his bland meal*.
2. *The lugubrious man is eating his bland meal*.
3. *The lugubrious man is eating his bland meal*.



'The lugubrious man is eating his bland meal.'

Exercise II: Constituency Tests



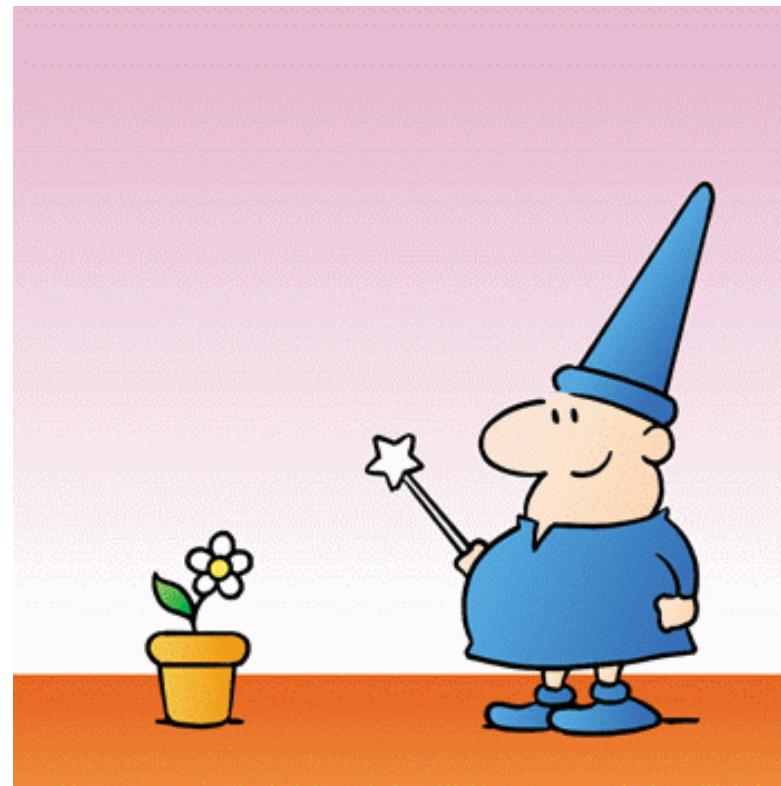
I thought this was common
knowledge

Use **constituency tests** to determine whether the underlined part is a **constituent** in each sentence:

1. The tragedy upset the entire family.
2. They hid in the cave.
3. The computer was very expensive.
4. The geese swam across the lake.
5. Merlin gave a book to his sister.
6. Arthur read a book about geology.
7. Morgana ate the nice meal of Arthur.

Careful with Constituency Tests

- if a given string **passes** a constituency test
 - → you can assume it is a constituent
- however, if a string **fails** a constituency test
 - → you cannot be certain that it is not a constituent
- sometimes, a test yields an **anomalous** result
 - *The student from Korea arrived.*
 - **clefting**: *It is from Korea that the student arrived.*
- this is why it is best to apply **several** constituency tests and not just rely on one



The Whole Picture

large phrases and
sentences



phrase structure rules

phrases



phrase structure rules

words

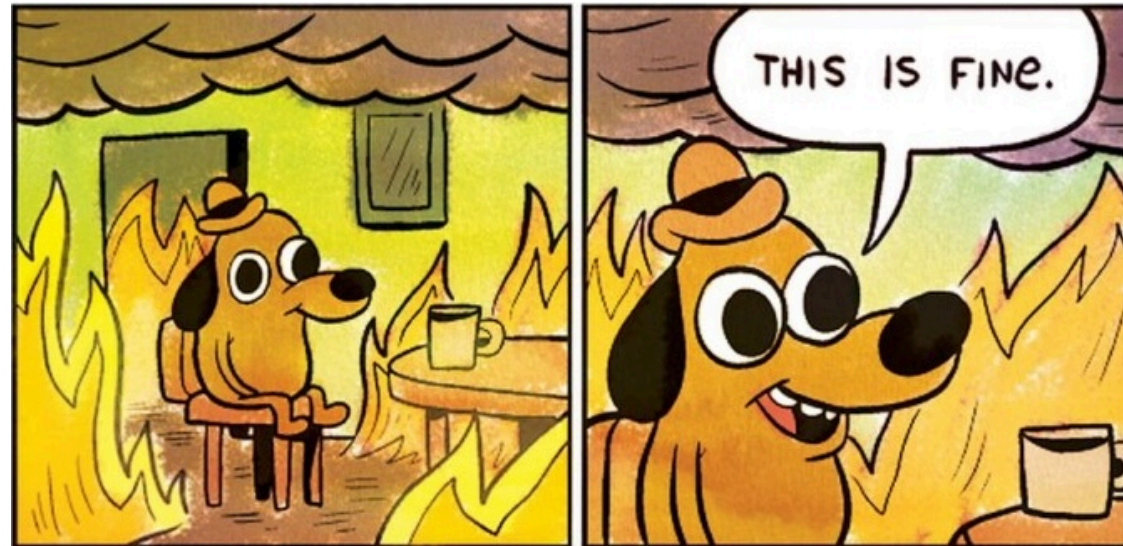


morphological rules

morphemes

Homework II is due this Friday (**Oct 02**)

reading Chapter 3: Constituency, Trees, and Rules from *Andrew Carnie's Syntax textbook*



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